

Python

Course Duration: 35Hrs

S. No	Content
1	Introduction of python • What is python • Who Developed python • Features of python • Applications of python • Setting up the Environment
2	Python Basic Syntax and Data Types • Input/output • Comments • Variables • Data types • Typecasting
3.	Operators in Python • Arithmetic • Assignment • Comparison • Logical • Identity • Membership • Bitwise operators

4.	Strings in Python • Creating • Formatting • Indexing • Slicing • String methods
5.	Lists in Python • Creating • Properties • Indexing • Slicing • Methods • Modifying lists
6.	Tuples in Python • Syntax • Properties • Indexing • Slicing • Methods
7.	Sets in Python • Syntax • Updating • Operations • Methods

8.	Dictionaries in Python • Syntax • Keys/values • Accessing • Methods
9	Python Conditional Statements • If statement • if-else • if-elif-else • Example Programs
10	Loops in Python • while • for • break • continue • pass • range
11.	Functions in Python • Creating • Calling • Arguments • Variables • Recursion • Anonymous Functions (Lambda)

12	Generators in Python • Creating • Using generators • Example program
13	Modules in Python • Creating • Importing • Built-in modules
14	Exceptions & Error Handling • Syntax errors • Logical errors • Try • Except • Finally
15	Classes & Objects • Creating • Attributes • Methods • Inheritance • Polymorphism

16	Date & Time • date • time • datetime modules
17	Regex in Python • re module • Common functions
18	Files in Python • Opening, Reading • Writing • Closing files
19	Exercise & Projects.